Rui Zhang

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Homepage: ruizhang.top **Scholar Page:** Google Scholar
Advisor: Prof. Zhenyu Wang **Research Interest:** NLP & Dialogue System

PARTICULARS

Education

2015.09 - Now South China University of Technology (SCUT) Guangzhou, China

- **Degree to Be Obtained:** Doctor of Software Engineering (Average Score: 88.27/100)
- **Honors and Awards:** National Scholarship for Doctoral Students (2018)

2011.09 - 2015.07 South China University of Technology (SCUT) Guangzhou, China

- **Degree and Major:** Bachelor of Software Engineering (GPA: 3.52/4.0)
- **Honors and Awards:** The First Prize Scholarship (2014)

Work Experience

2014.07 - 2014.09 Kingsoft WPS Office Software

- C++ Development Engineer (Intern)

PUBLICATIONS

1. **Predicting Emotion Reactions for Human-Computer Conversation: A Variational Approach**
   Rui Zhang, Zhenyu Wang✉, Zhenhua Huang, Li Li and Mengdan Zheng.
   In *IEEE Transactions on Human-Machine Systems*, 2020.
2. **Emotional Text Generation Based on Cross-Domain Sentiment Transfer**
   Rui Zhang, Zhenyu Wang✉, Kai Yin and Zhenhua Huang.
   In *IEEE Access*, 2019.
3. **Learning to Converse Emotionally Like Humans: A Conditional Variational Approach**
   Rui Zhang and Zhenyu Wang✉.
   In *CCF International Conference on Natural Language Processing and Chinese Computing (NLPCC)*, 2018.
4. **Building Emotional Conversation Systems using Multi-task Seq2Seq Learning**
   Rui Zhang, Zhenyu Wang✉ and Dongcheng Mai.
   In *CCF International Conference on Natural Language Processing and Chinese Computing (NLPCC)*, 2017.
5. **Melodic Phrase Attention Network for Symbolic Data-based Music Genre Classification (Student Abstract)**
   Li Li, **Rui Zhang** and Zhenyu Wang✉.
   Proceedings of *Student Abstract Program of the AAAI Conference on Artificial Intelligence (AAAI)*, 2021.
6. **Learning Bi-directional Social Influence in Information Cascades using Graph Sequence Attention Networks**
   Zhenhua Huang , Zhenyu Wang✉ , **Rui Zhang**, Yangyang Zhao and Fadong Zheng.
   In *the Web Conference(WWW)*, 2020.
7. **Cascade2vec: Learning Dynamic Cascade Representation by Recurrent Graph Neural Networks**
   Zhenhua Huang, Zhenyu Wang✉, and **Rui Zhang**.
   In *IEEE Access*, 2019.
8. **Network2Vec: Learning Node Representation Based on Space Mapping in Networks**
   Zhenhua Huang, Zhenyu Wang✉, **Rui Zhang**, Yangyang Zhao, Xiaohui Xie, and Sharad Mehrotra.
   In *Proceedings of the IEEE ICDM Workshop on Social Computing (ICDM)*, 2019.
9. **MMCRD: An Effective Algorithm for Deploying Monitoring Point on Social Network**
   Zehao Guo, Zhenyu Wang✉, and **Rui Zhang**.
   In *China Conference on Knowledge Graph and Semantic Computing (CCKS)*, 2018.

10. **Dynamic Reward-based Dueling Deep Dyna-Q: Robust Policy Learning in Noisy Environments**
    Yangyang Zhao, Zhenyu Wang✉, Kai Yin, **Rui Zhang**, Zhenhua Huang.
    In *AAAI Conference on Artificial Intelligence (AAAI)*, 2020.
11. **TRAIN: A Deep Topic and Role-Aware Information Diffusion Model in Social Networks**
    Zhenhua Huang, Zhenyu Wang✉, Dongfa Zheng, **Rui Zhang**
    In *Chinese National Conference on Social Media Processing (SMP)*, 2020.
12. **Music Auto-tagging Algorithm Based on Deep Analysis on Labels**
    Zhenyu Wang✉, **Rui Zhang**, Yuxuan Gao and Yongle Xiao.
    In *Journal of South China University of Technology (Natural Science Edition)*, 2019.
13. **A Survey on Task-Oriented Dialogue Systems**
    Yangyang Zhao, Zhenyu Wang✉, Pei Wang, Tian Yang, **Rui Zhang**, Kai Yin
    In *Chinese Journal of Computers,* 2019.

PROJECTS

1. **Intelligent Music Search and Recommendation Platform based on Big Data Analysis** **Core Member**
   Science and Technology Planning Project of Guangzhou (No.201802010025)
2. **Industrialized Mobile Application of Big Data Analysis and Recommendation Platform** **Core Member**
   Application Technology R&D Project of Guangdong Province (No.2015B010131003)
3. **Big Data Management and Analysis Technology of Mobile Internet User Behavior** **Member**
   Collaborative Project of Industry, University and Research in Guangzhou (No.201604010017)
4. **IoT Security Service Framework based on Trusted Computing** **Core Member**
   Development and Guidance Project of Guangdong High-tech Zone (No.2012B010900053)

INTELLECTUAL PROPERTIES

1. **Patent: Automatic Music Labeling Method based on Deep Label Analysis**
   The third author. Status: Accepted. (No.201910047030.0)
2. **Software Copyright: Secure Transmission Middleware System of IoT for Android Devices**
   The second author. Status: Registered. (No.2014SR069341)
3. **Software Copyright: Optional Secure Transmission Middleware System for IoT Applications**
   The fourth author. Status: Registered. (No.2014SR069580)

AWARDS

1. Participated in the "Emotional Dialogue Generation" technical evaluation of the NLPCC2017 conference, and won **the 3rd place** among 40 participating teams. Our multi-task emotional dialogue generation system (MECS) won **the 1st place** in the completely generation-based systems (non-retrieval based).
2. Participated in the 2016 "Magic Mirror Cup Internet Financial Data Application - Data Product Development" competition. The cloud-based big data monitoring system developed by the team won **the national 6th place (winning prize)** among 58 participating teams.

PERSONAL SKILLS

1. Proficiency in Python/C++/Java dev. , Data Struture and Pycharm/Eclipse/Vim IDEs.
2. Depth knowledge of Deep Learning, expertise in Tensorflow/Keras/Pytorch frameworks.
3. Familiar with Linux, experienced in common command and shell.
4. Mastered in English reading/writing/communication

张睿

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谷歌学术主页: Google Scholar
导师: 王振宇 教授
研究领域: 自然语言处理 、对话系统

个人履历

教育背景

2015.09 - Now 华南理工大学 (SCUT)

- 即将取得的学位: 软件工程学术型博士 (课程成绩: 88.27/100)
- 荣誉与奖项: 于 2018 年获得博士生国家奖学金

2011.09 - 2015.07 华南理工大学 (SCUT)

- 学位与专业: 软件工程工学学士 (绩点: 3.52/4.0)
- 荣誉与奖项: 于 2014 年获得华南理工大学校级一等奖学金，同年获保送直攻博士资格

工作经历

2014.07 - 2014.09 金山 WPS 办公软件

- C++ 开发工程师 (实习)

已发表的论文

1. **Predicting Emotion Reactions for Human-Computer Conversation: A Variational Approach**
   Rui Zhang, Zhenyu Wang✉, Zhenhua Huang, Li Li and Mengdan Zheng.
   In *IEEE Transactions on Human-Machine Systems*, 2020.（CCF-B 类/中科院 2 区/JCR Q1 期刊，IF: 3.374；SCI 收录）
2. **Emotional Text Generation Based on Cross-Domain Sentiment Transfer**
   Rui Zhang, Zhenyu Wang✉, Kai Yin and Zhenhua Huang.
   In *IEEE Access*, 2019.（中科院 2 区/JCR Q1 期刊，IF: 4.098；SCI 收录）
3. **Learning to Converse Emotionally Like Humans: A Conditional Variational Approach**
   Rui Zhang and Zhenyu Wang✉.
   In *CCF International Conference on Natural Language Processing and Chinese Computing (NLPCC)*, 2018.（EI 会议）
4. **Building Emotional Conversation Systems using Multi-task Seq2Seq Learning**
   Rui Zhang, Zhenyu Wang✉ and Dongcheng Mai.
   In *CCF International Conference on Natural Language Processing and Chinese Computing (NLPCC)*, 2017.（EI 会议）
5. **Melodic Phrase Attention Network for Symbolic Data-based Music Genre Classification (Student Abstract)**
   Li Li, **Rui Zhang** and Zhenyu Wang✉.
   Proceedings of *Student Abstract Program of the AAAI Conference on Artificial Intelligence (AAAI)*, 2021.（CCF-B 会议）
6. **Learning Bi-directional Social Influence in Information Cascades using Graph Sequence Attention Networks**
   Zhenhua Huang , Zhenyu Wang✉ , **Rui Zhang**, Yangyang Zhao and Fadong Zheng.
   In *the Web Conference(WWW)*, 2020.（CCF-A 会议）
7. **Cascade2vec: Learning Dynamic Cascade Representation by Recurrent Graph Neural Networks**
   Zhenhua Huang, Zhenyu Wang✉, and **Rui Zhang**.
   In *IEEE Access*, 2019.（中科院 2 区/JCR Q1 期刊，IF: 4.098；SCI 收录）
8. **Network2Vec: Learning Node Representation Based on Space Mapping in Networks**
   Zhenhua Huang, Zhenyu Wang✉, **Rui Zhang**, Yangyang Zhao, Xiaohui Xie, and Sharad Mehrotra.
   In *Proceedings of the IEEE ICDM Workshop on Social Computing (ICDM)*, 2019.（CCF-B 会议）
9. **MMCRD: An Effective Algorithm for Deploying Monitoring Point on Social Network**
   Zehao Guo, Zhenyu Wang✉, and **Rui Zhang**.
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10. **Dynamic Reward-based Dueling Deep Dyna-Q: Robust Policy Learning in Noisy Environments**
    Yangyang Zhao, Zhenyu Wang✉, Kai Yin, **Rui Zhang**, Zhenhua Huang.
    In *AAAI Conference on Artificial Intelligence (AAAI)*, 2020.（CCF-B 会议）
11. **TRAIN: A Deep Topic and Role-Aware Information Diffusion Model in Social Networks**
    Zhenhua Huang, Zhenyu Wang✉, Dongfa Zheng, **Rui Zhang**
    In *Chinese National Conference on Social Media Processing (SMP)*, 2020.（EI 会议）
12. **基于标签深度分析的音乐自动标注算法.Music Auto-tagging Algorithm Based on Deep Analysis on Labels**
    Zhenyu Wang✉, **Rui Zhang**, Yuxuan Gao and Yongle Xiao.
    In *华南理工大学学报(自然科学版). Journal of South China University of Technology (Natural Science Edition)*, 2019.
13. **任务型对话系统研究综述. A Survey on Task-Oriented Dialogue Systems**
    Yangyang Zhao, Zhenyu Wang✉, Pei Wang, Tian Yang, **Rui Zhang**, Kai Yin
    In *计算机学报. Chinese Journal of Computers,* 2019.

项目经历

1. **基于大数据分析的移动音乐智能搜索与推荐平台（No.201802010025）** 主要参与人
   广州市科技计划项目（产业技术重大攻关计划-现代产业技术专题），2017.10~2020.9，已结题
2. **面向移动互联网用户大数据分析及推荐平台的产业化应用（No.2015B010131003）** 主要参与人
   广东省应用型科技研发专项资金项目重点项目，2015.9~2017.8，已结题
3. **移动互联网用户行为大数据管理与分析技术研发及产业化（No.201604010017）** 参与人
   广州市产学研协同创新重大专项，2015.5~2017.4，已结题
4. **基于可信计算的物联网安全服务框架（No.2012B010900053）** 主要参与人
   广东省高新区发展引导专项（攻关及产业化类），已结题

已申请的知识产权

1. **发明专利：基于标签深度分析的音乐自动标注方法（No.201910047030.0）**
   发明人：导师第一，本人第三；状态：已受理
2. **软件著作权：面向 Android 设备的物联网安全传输服务中间件系统 v1.0（No.2014SR069341）**
   发明人：导师第一，本人第二；状态：已登记
3. **软件著作权：面向物联网应用的可选安全传输模式中间件系统 v1.0（No.2014SR069580）**
   发明人：导师第一，本人第四；状态：已登记

技术评测/竞赛获奖情况

1. 参加 NLPCC2017 会议“Emotional Conversation Generation（情感对话生成）”技术评测，所提出的基于多路情感对话生成系统（MECS）在参赛的 40 支队伍中取得第 3 名的成绩。其中，在完全基于深度文本生成（非检索）的系统中取得了第 1 名的成绩。
2. 参加 2016 年“魔镜杯互联网金融数据应用——数据产品开发”大赛，所开发的基于云的大数据监控系统在参赛的 58 支队伍中取得全国第 6 名（优胜奖）的成绩。

个人技能

1. 熟练使用 Python，熟悉 C++/Java/Matlab，熟悉 Pycharm/Eclipse/Vim 等开发环境
2. 熟悉深度学习技术，熟练使用 TensorFlow/Keras/Pytorch 框架，了解 Theano/PaddlePaddle 框架
3. 熟悉 Linux 操作系统，熟练使用常用命令和 Shell 脚本
4. 良好的英语阅读/写作/交际能力